

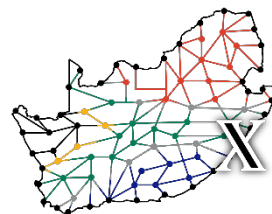
DEEP LEARNING INDABAX SOUTH AFRICA 2019



14 - 17 APRIL 2019

UNITE Building
University of KwaZulu-Natal
Durban

DEEP LEARNING INDABA $\times$ SOUTH AFRICA 2019



ABOUT US

The Deep Learning Indaba $\times$ is a locally-organised meeting that helps ensure that knowledge and capacity in machine learning are spread more widely across the continent. The Indaba $\times$ is a means with which to experiment with the ways in which we strengthen African machine learning and its local communities, allowing more people to contribute to the conversation around machine learning and artificial intelligence. Indaba $\times$ events are local versions of the Deep Learning Indaba, which will be held this year in Nairobi, Kenya.

LIST OF ORGANISERS

AMIRA ABBAS

University of KwaZulu-Natal

CHRISTOPHER CURRIN

University of Cape Town

ANBAN PILLAY

University of KwaZulu-Natal

JAMES ALLINGHAM

Wolfram Research

STEVEN JAMES

University of the Witwatersrand

MARIA SCHULD

University of KwaZulu-Natal

ANNIKA BRUNDYN

University of Cape Town

SASHA NAIDOO

Nedbank

SICELUKWANDA ZWANE

University of the Witwatersrand

ALEX CONWAY

NumberBoost

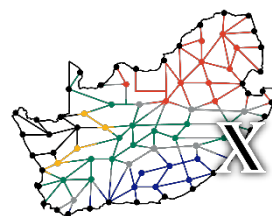
ANDREW PASKARAMOORTHY

University of the Witwatersrand

VENUE INFO

All talks take place at the UNITE Building, University of KwaZulu-Natal. The discussion, keynote and research talks take place in the Lecture Theatre. Foundation talks occur in Rooms 4 and 5, and industry, ethics and policy talks take place in Rooms 1, 2 and 3. The opening event will take place at the PYC Yacht Club, 3 Maritime Pl, Harbour, Durban.

DEEP LEARNING INDABAX SOUTH AFRICA 2019



TRACKS

KEYNOTE



FOUNDATIONS



RESEARCH



INDUSTRY, ETHICS & POLICY



HACKATHON



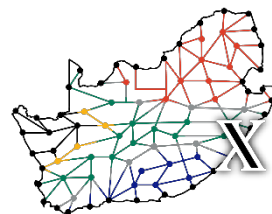
UNCONFERENCE



OVERVIEW

14 April SUNDAY		15 April MONDAY		16 April TUESDAY			17 April WEDNESDAY	
08h30		Opening Remarks (15 mins)						
09h00		Discussion: Africa in AI						
09h30								
10h00								
10h30		Tea Break		Tea Break				
11h00								
11h30								
12h00								
12h30								
13h00		Lunch		Lunch				
13h30								
14h00								
14h30								
15h00								
15h30	Tea Break		Tea Break			Prizegiving and Closing		
16h00	Registration							
16h30								
17h00								
17h30								
18h00	Opening Event		Poster Session					
18h30								
19h00								
19h30								
20h00								

DEEP LEARNING INDABAX SOUTH AFRICA 2019



DAY 0

Sunday, 14 April

ATTENDEE REGISTRATION

UNITE Building, University of KwaZulu-Natal

15h00 - 18h00

OPENING EVENT

PYC Yacht Club, 3 Maritime Place, Harbour, Durban

18h00 - 22h00

DAY 1

Monday, 15 April

OPENING REMARKS

08h30 - 08h45

OPENING DISCUSSION: AFRICA IN AI

Tshilidzi Marwala, Benjamin Rosman, Justine Nasejje, Pelonomi Moiloa, Jade Abbott, Sicelukwanda Zwane & Chris Currin

08h45 - 10h30

TEA BREAK

10h30 - 11h00

KEYNOTE I

Tshilidzi Marwala: On the Rationality of Artificial Intelligence Machines

11h00 - 13h00

LUNCH

13h00 - 14h00

FOUNDATIONS I

Kerryn Gammie: Animated Intro to Machine Learning

14h00 - 15h30

RESEARCH I

Ndivhuwo Makondo: Machine Learning For Robot Control

14h00 - 14h30

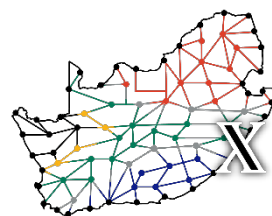
Ashley Gritzman: Will Capsule Networks Replace CNNs?

14h30 - 15h00

Terence van Zyl: Random Neural Networks

15h00 - 15h30

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INDUSTRY, ETHICS & POLICY I

<i>Nick Bradshaw:</i> Community Development as an Avenue to Career Growth	14h00 - 14h20
<i>Alex Conway:</i> Sebenz.ai - Creating Jobs in South Africa using Smartphone Data Labeling	14h20 - 15h00
<i>Jade Abbott:</i> Machine Learning in Real Life	15h00 - 15h30

TEA BREAK

15h30 - 16h00

FOUNDATIONS II

<i>Richard Klein:</i> Introduction to Deep Learning	16h00 - 17h30
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RESEARCH II

<i>Emlī-Mari Nel:</i> Ticker: An Adaptive Single-Switch Text Entry Method for Visually Impaired Users	16h00 - 16h30
<i>Sebastian Bodenstein & Taliesin Beynon:</i> Deep Learning Vs The Brain	16h30 - 17h00
<i>James Allingham:</i> Missing Data Imputation	17h00 - 17h30

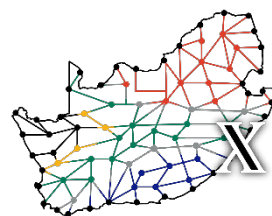
INDUSTRY, ETHICS & POLICY II

<i>Nick Pentreath:</i> The IBM Developer Model Asset Exchange - Deep Learning For Everyone	16h00 - 16h30
<i>Stuart Reid:</i> The Temporal Turing Test: Sampling Real-Valued Multivariate Time Series from Deep Generative Models	16h30 - 17h00
<i>Jan Krynauw & Zander Lee:</i> Humans in the Loop - New Ways of Working in a Deep Learning World	17h00 - 17h30

POSTER SESSION

Hosted in the Foyer. Snacks and drinks provided	17h30- 20h00
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DAY 2

Tuesday, 16 April

KEYNOTE II

Andrew Saxe: High-Dimensional Dynamics of Generalization Error in Neural Networks: Insights from the Linear Case

08h30 - 10h30

TEA BREAK

10h30 - 11h00

FOUNDATIONS III

Wiebke Toussaint: Unsupervised Machine Learning 101

11h00 - 12h00

Edgar Jembere: Introduction to Natural Language Processing

12h00 - 13h00

RESEARCH III

Justine Nasejje: Using Conditional Inference Forests to Model Time to Event Data

11h00 - 11h30

Willie Brink: Visual-Semantic Embeddings: An Introduction and Applications

11h30 - 12h15

Benjamin Rosman: Exploiting Structure for Accelerating Reinforcement Learning

12h15 - 13h00

INDUSTRY, ETHICS & POLICY III

Jacques Ludik: Digital Transformation Journey in Operationalizing AI Solutions to Help Businesses Thrive in the Smart Technology Era

11h00 - 11h30

Quentin Williams: AI + Humans in Business: Progeny or Friendship

11h30 - 12h00

Pelonomi Moiloa: Protecting Machines from us - Ethics and Bias

12h00 - 12h30

Matthew Adendorff: Civic Technology and AI - Meaningful Opportunities and Daunting Biases

12h30 - 13h00

LUNCH

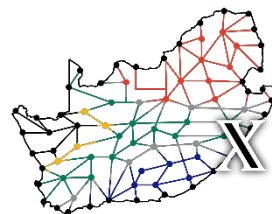
13h00 - 14h00

FOUNDATIONS IV

Jonathan Shock: A Guided Tour through the Landscape of Reinforcement Learning

14h00 - 15h30

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RESEARCH IV

Christopher Cleghorn: Black-Box Optimization and its Role in Machine Learning

14h00 - 14h45

Maria Schuld: Machine Learning with Quantum Computers

14h45 - 15h30

INDUSTRY, ETHICS & POLICY IV

John Poole: Centre for High Performance Computing: Resources for Deep Learning at the CHPC

14h00 - 14h30

Michelle Hoogenhout: Growing young talent: Selecting and training the next generation of data scientists

14h30 - 15h00

Dario Fannucchi: A Tour of Machine Learning in the African Business Context: 5 Years of Awesome Ideas, Dramatic Flops and Big Wins

15h00 - 15h30

TEA BREAK

15h30 - 16h00

KEYNOTE III

Bruce Bassett: An AI Play in 3 Parts: Jacuzzis, Research and AGI

16h00 - 17h30

HACKATHON OPENING

Hosted in the Lecture Theatre and LAN until 01h00

17h30 - Onward

DAY 3

Wednesday, 17 April

HACKATHON CONTINUES

Hosted in the LAN and Rooms 1, 2 and 3

08h30 - 15h00

UNCONFERENCE

Hosted in the Lecture Theatre and Rooms 1, 2 and 3

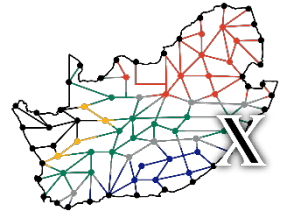
08h30 - 15h00

PRIZE-GIVING AND CLOSING REMARKS

Poster winner and hackathon prizes announced

15h00 - 15h30

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